Experiment No. 2 – Sample Application Development

Aim

To develop a simple application and understand the basic software development process.

Objectives

- To write and execute a simple application.
- To understand program design and implementation.
- To test the application with different inputs.
- To prepare the application for version control and DevOps practices.

Theory

Application development is the process of designing, coding, testing, and deploying software applications to solve specific problems. In DevOps, application development is the first stage of the software delivery pipeline. The developed application is later integrated with version control, continuous integration, testing, and deployment tools.

Problem Statement

Develop a simple **Student Grade Calculator** application that accepts a student's name and marks, calculates the grade, and displays the result.

Algorithm

1. Start.
2. Input student name.
3. Input student marks.
4. Check the marks range.
5. Assign the corresponding grade.
6. Display student name, marks, and grade.
7. Stop.

Program Code (Python)

```
name = input("Enter Student Name: ")
marks = float(input("Enter Marks: "))

if marks >= 90:
    grade = "A+"
elif marks >= 80:
    grade = "A"
elif marks >= 70:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"

print("\nStudent Name:", name)
print("Marks:", marks)
print("Grade:", grade)
```

Procedure

1. Open a Python IDE or terminal.
2. Create a file named grade_calculator.py.
3. Enter the given program code.
4. Save the file.
5. Execute the program using Python.
6. Enter the student name and marks.
7. Observe the generated grade.

Sample Input

Enter Student Name: Ajit

Enter Marks: 85

Sample Output

Student Name: Ajit

Marks: 85.0

Grade: A

Test Cases

Marks	Expected Grade
95	A+
85	A
75	B
65	C
50	Fail